Dense Flow from Adaptive Correspondence

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Abstract

Optical flow requires reliable matching from sufficient visual context while estimating precise motion at every pixel. State-of-the-art (SOTA) methods such as WAFT match on fixed grids and then upsample flow, without explicitly linking how visual evidence is gathered for matching to how motion is distributed to each pixel. Our key insight is that pixels likely to move together can share visual evidence for matching and draw on the same motion estimates; this consistent visual grounding lets correspondence operate at a coarser scale than the flow. We realize this in *Flow from Adaptive Token Estimation* (FATE), which learns adaptive correspondence tokens whose supports spread within coherent regions to gather context and sharpen at motion boundaries to keep flow precise. Shared soft pixel-to-token associations pool features for matching and combine coarse token motions into precise pixel flow. Flow supervision jointly trains grouping, matching, and flow estimation without segmentation labels, enabling the model to anticipate which pixels are likely to move together from appearance alone in unseen images. Trained on FlyingChairs/FlyingThings3D, FATE achieves SOTA *zero-shot* accuracy on Sintel Final and KITTI, a critical setting given scarce real-world flow labels. For visible pixels, FATE improves accuracy within coherent regions, and more so near motion boundaries. Its largest gains occur where direct matches are unavailable: Occluded or out-of-frame pixels account for roughly 10% of valid pixels but 50% of WAFT's total error. Experiments show that learning adaptive correspondence through pixel-to-token grouping enables both reliable matching and precise pixel-level flow, and improves generalization to unseen real scenes.

1 Introduction

Visual correspondence asks which elements in two images refer to the same scene content. We study two-image optical flow, which estimates the displacement from each pixel in the first image to its corresponding location in the second, even when that location is occluded or outside the image [1, 2].

Optical flow must meet two demands: finding reliable matches using sufficient context and estimating precise motion at every pixel. A distinctive corner is easy to match across two images, making its displacement straightforward to estimate. A pixel on a white wall, however, could match many white pixels in the second image, making its displacement hard to pinpoint. Yet the wall itself may be easier to identify. Similarly, when part of an object becomes occluded or leaves the image, its motion may be inferred from the parts still visible. In both cases, matching coarser structures can help estimate the precise motion of pixels that cannot be reliably matched on their own.

These two demands pull in different directions: Reliable matching may require larger image regions to resolve ambiguity, while precise motion localization requires smaller regions to distinguish the movements of individual pixels, especially those on opposite sides of motion boundaries [3, 4].

Modern flow methods largely address these demands in two stages. They first combine information from larger image regions to build richer features for more reliable matching at fixed grid locations

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[5–8], then recover precise pixel flow through learned upsampling [5, 8]. Ideally, the pixels that supply evidence for a match should also receive its motion. In these two-stage methods, however, the image region each grid feature summarizes and the upsampling weights that distribute its motion are linked only implicitly, through end-to-end training. This raises the question of whether this implicit link generalizes from synthetic training data to unseen real scenes.

Our key insight is to reconcile reliable matching with precise localization by separating correspondence scale from flow resolution, while using the same pixel grouping for matching and for distributing motion to pixels. **1) Correspondence should adapt** to image structure, matching distinctive visual entities through spatial supports that make them reliably identifiable. Motion from these matches can then be combined to recover precise flow at every pixel. **2) The same grouping should govern** which pixels supply evidence for each match and how the resulting motion is distributed back to them. This ties precise localization to the same evidence used for reliable matching. **3) Grouping should anticipate common fate from visual appearance.** The Gestalt principle of *common fate* holds that elements moving together are perceived as a group [9, 10]. Yet correspondence requires deciding what belongs together *before* knowing how it moves. Prior studies have used motion to learn appearance-based object segmentation [11, 12]. We instead use flow supervision to *learn appearance-based grouping for correspondence*, allowing the model to predict these groups in unseen images for more reliable and precise flow estimation.

We realize these ideas in FATE, short for *Flow from Adaptive Token Estimation*. **1) Adaptive correspondence tokens** enable coarse-scale matching while retaining pixel-level flow prediction. Each token is anchored to a coarse grid, but its spatial support adapts to image features. As Fig. 1 shows, token supports spread within coherent regions and sharpen at motion boundaries, following regions of common motion rather than fixed grid cells. Local attention enriches token descriptors with broader context, while iterative matching estimates token motion [8]. **2) Pixel-to-token grouping** connects matching and localization through a shared differentiable association matrix: Normalized over pixels, it pools features into token descriptors; normalized over tokens, it combines token motions into precise pixel flow. Each token estimates one motion vector, while pixels combine token motions to preserve distinct flows across boundaries. **3) Flow supervision teaches appearance-based grouping.** Pixel-to-token associations, token matching, and dense flow estimation are learned jointly, without segmentation labels. At test time, the model predicts these associations from appearance alone, enabling more reliable and precise flow estimation in unseen images.

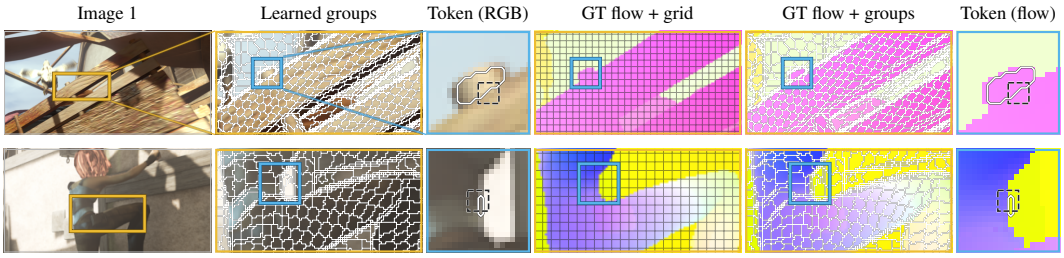


Figure 1: **FATE learns correspondence tokens whose spatial support adapts to image structure and motion boundaries.** Token supports spread within coherent regions and sharpen at motion boundaries, following common motion rather than fixed grid cells. Learned groups trace slat edges and a bent body outline, while fixed cells can straddle distinct motions. The same soft pixel-to-token associations pool features for matching and combine token motions into dense flow. White contours show dominant assignments and selected token supports; dashed squares mark nominal grid cells. Selected supports appear against RGB images and ground-truth (GT) flow in the token panels. Gold and blue boxes mark successive enlargements, with matching RGB and GT flow crops.

These ideas deliver stronger zero-shot generalization, a critical need in optical flow. Dense flow labels are difficult to obtain in real scenes, so supervised models are often trained on synthetic images and must transfer to unseen domains without fine-tuning. While visual appearance may vary across domains, cues to which pixels are likely to move together may still transfer. Trained on FlyingChairs and FlyingThings3D [13, 14], FATE achieves endpoint errors of 2.10 on Sintel Final and 3.11 on KITTI, the lowest among the published methods in our zero-shot comparison [15, 16]. Its largest gains occur where direct matches are unavailable: Occluded or out-of-frame pixels account for only 6.4–13.9% of valid pixels, yet contribute 41.9–52.7% of WAFT's total endpoint error. At the same

number of matching locations, FATE outperforms fixed-grid alternatives and benefits more from local attention; its tokens also form a better basis for dense flow, preserving distinct motions that fixed grid cells average away. With four refinement steps, it also outperforms five-step WAFT at lower cost.

In summary, we make three contributions. **1) Adaptive correspondence tokens** improve flow accuracy with fewer matching locations through learned spatial supports. **2) Shared pixel-to-token associations** provide consistent visual grounding for matching and dense flow estimation. **3) Strong zero-shot generalization** is demonstrated across benchmarks and supported by controlled analyses. These advances simultaneously enable reliable matching and precise pixel-level motion estimation, while generalizing effectively to unseen real-world scenes. Project website: FATE.

2 Related Work

Spatial grouping in classical optical flow. Classical optical flow uses spatial support to infer motion from ambiguous local measurements. Lucas–Kanade [2] pools motion constraints within neighborhoods, while Horn–Schunck [1] couples neighboring estimates through smoothness. Robust estimation [3] accommodates competing motions. Ren [4] uses contour-based soft affinities and figure-ground organization to define spatially adaptive support for local affine motion estimation near boundaries. FATE learns spatial support jointly with matching features from flow supervision, using the resulting groups for correspondence and dense flow estimation.

Common fate and motion-guided grouping. The Gestalt principle of common fate holds that elements moving together are perceived as a group [9, 10]. AMD [11] jointly learns appearance-based segmentation and segment motion through video reconstruction, pooling motion features to predict segment flow. RCF [12] relaxes common fate by fitting flow with segment motion and within-segment residuals, then refines grouping from appearance. These methods learn object segmentation from motion, whereas FATE learns grouping for correspondence and dense flow. Its soft associations need not follow object boundaries or impose uniform flow within a token.

Correspondence in deep flow networks. Modern two-frame networks largely match features on regular grids. Learned pyramids and cost volumes [5, 17], global matching and cost-memory encoding [6, 7], and pretrained representations [18, 19] strengthen features for matching. Recurrent estimation and learned motion priors [20, 21] improve flow estimation. Source-image affinities aggregate motion features [22], while feature similarity guides flow propagation [6]. Multiscale processing and learned upsampling recover spatial detail [5, 8, 23, 24]. In these methods, pixel support for matching and the weights that distribute motion to pixels are linked only implicitly. Building on iterative matching [8], FATE learns adaptive correspondence tokens whose shared pixel-to-token associations pool features for matching and combine token motions into dense flow.

Superpixels and differentiable grouping. Superpixel methods partition images into compact regions following appearance boundaries [25, 26]. Learned variants make grouping differentiable and adaptable to downstream tasks [27, 28]. Unlike image segmentation, grouping in FATE serves correspondence and dense motion estimation. Flow supervision jointly learns the associations and matching features, without requiring pixel-group labels.

3 FATE: Flow from Adaptive Token Estimation

Overview. Given two images, FATE estimates dense optical flow by matching adaptive correspondence tokens at a coarser resolution, then combining their motions into pixel-level flow (Fig. 2). The model has three components. **1) Adaptive correspondence tokens** are anchored to points on a coarse grid, but their spatial supports are learned from image features. **2) Shared pixel-to-token associations** pool each image's dense features into token descriptors, enriched by local attention for broader matching context. Iterative matching estimates token motion, which the same associations combine into pixel flow in the first image. Since each pixel combines motions from several tokens, pixels within a token's support need not share identical flow. **3) Flow-supervised grouping** jointly learns associations, matching, and flow estimation from ground-truth pixel flow, without pixel grouping labels. At test time, the model anticipates which pixels are likely to move together from appearance alone, enabling reliable matching and precise pixel flow in unseen scenes.

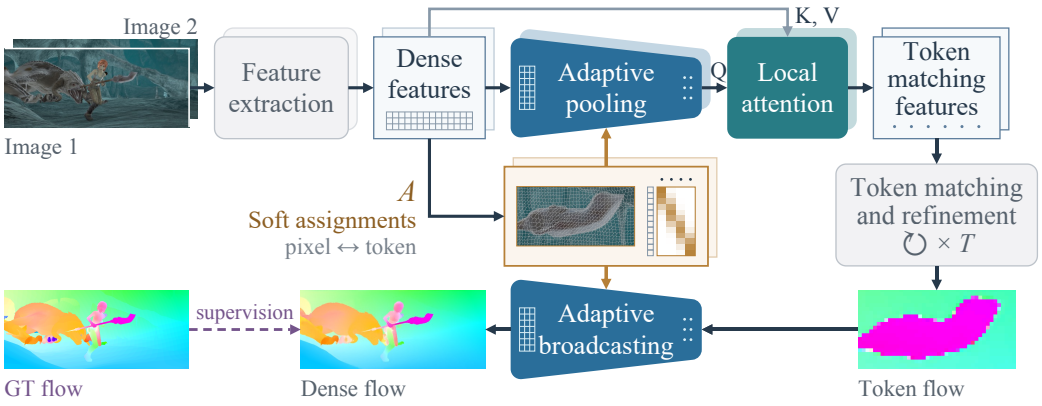


Figure 2: **Shared pixel-to-token associations connect token matching and dense flow estimation.** Learned associations A pool dense image features into adaptive correspondence tokens (center), enriched with nearby context through local attention before iterative matching across images (right). The same associations combine estimated token motions into dense pixel flow (bottom), allowing each pixel to draw on multiple tokens. Image 2 (lighter branch) has its own associations but shares network weights with Image 1. Dashed arrows indicate ground-truth flow supervision (Fig. 4).

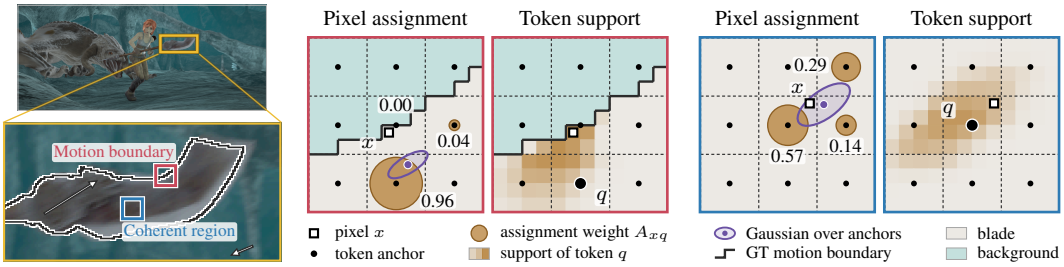


Figure 3: **Learned pixel-to-token associations adapt token support to motion structure.** Left: the original image and an enlarged view of the moving blade, with two marked regions: a motion boundary and a coherent region. Arrows indicate ground-truth motion. Center and right: Each region is shown from two perspectives. Pixel assignment shows how pixel x associates with nearby tokens (a row of A); token support shows the spatial support of token q (a column of A). Dashed grid lines mark the coarse token lattice, black dots mark token anchors, and squares mark pixels. Brown disks represent individual assignment weights A_{xq}, with size indicating weight; purple ellipses show predicted Gaussians, and brown shading shows token support. At the motion boundary, x favors a blade token across grid-cell borders, allowing its support to align with the boundary. Within the coherent region, x distributes its associations across several nearby tokens.

Adaptive correspondence tokens. For an image with N pixels indexed by x and M tokens indexed by q, we learn a pixel-to-token association matrix $A \in \mathbb{R}^{N \times M}$ of soft assignments, where $A_{xq} \geq 0$ measures how strongly pixel x is associated with token q. Each row describes a pixel's associations with tokens, while each column describes a token's spatial support.

Predicting associations from appearance. Each pixel associates with the 3×3 token anchors around its grid cell. Rather than predicting nine independent scores, a compact head predicts a two-dimensional Gaussian over these anchors from dense image features. Its center determines which nearby tokens the pixel favors, while its shape controls how widely the associations spread. Evaluating the Gaussian at the candidate anchors and normalizing over them gives the corresponding entries of A (Appendix G). As Fig. 3 illustrates, the resulting token supports can follow motion boundaries rather than fixed cell borders, while spreading more broadly within coherent regions.

Shared pixel-to-token associations. The same learned associations govern how pixels support their token matching and how token motions are combined into dense pixel flow (Fig. 4).

1) Pooling and broadcasting. The association matrix A connects pixel and token spaces in two ways: Normalized over pixels (columns of A), it pools features into token descriptors; normalized over tokens (rows of A), it combines token motions into pixel flow. Let $D(B) = \mathrm{diag}(B\mathbf{1})$ denote the diagonal degree matrix of row sums. Omitting the dependence on A, we have two operators:

$$\text{pixel-to-token feature pooling:} \quad P_q = D(A^\top)^{-1}A^\top, \tag{1}$$

$$\text{token-to-pixel flow broadcasting:} \quad P_x = D(A)^{-1}A. \tag{2}$$

The shared associations provide consistent visual grounding for matching and pixel flow.

2) Token features with local attention. Reliable matching requires both features pooled from each token's support and broader context to resolve ambiguity. Local attention contextualizes each token's pooled features with surrounding dense features, sharing network weights for both images.

3) Iterative flow refinement and broadcasting. Let F_q and F_x denote token and pixel flow from the first image to the second. Following the iterative matching framework of WAFT [8], each refinement step samples the second image's token features at current match locations and updates the token flow. Both images' associations are computed once and reused throughout refinement. At step t, the first image's broadcasting operator produces dense pixel flow:

$$F_x^{(t)} = P_x F_q^{(t)}. \tag{3}$$

Each token predicts one motion vector, but each pixel combines motions from multiple tokens, allowing distinct pixel flow estimates within the same token's spatial support.

Flow-supervised grouping. We jointly learn pixel-to-token associations and token-flow prediction from ground-truth pixel flow, since no annotation specifies which pixels should be grouped for correspondence (Fig. 4). Three complementary losses supervise token matching, pixel-to-token associations, and preservation of ground-truth pixel motion through pooling and reconstruction. Each has the form $\mathcal{L}(X \mid Y)$, training X with Y given.

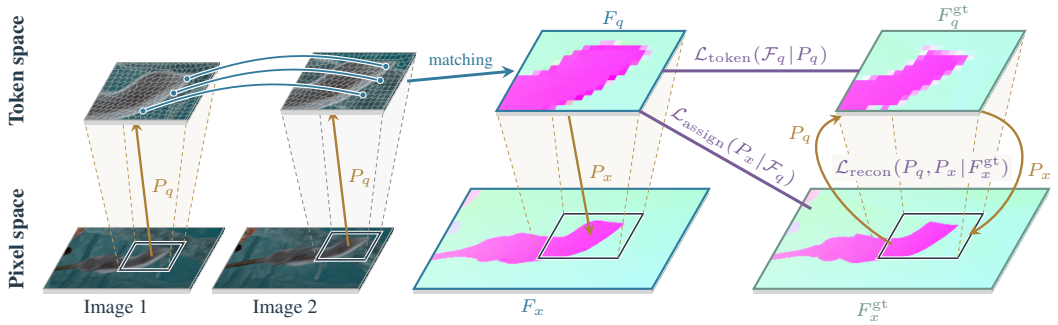


Figure 4: **Decoupling token-level matching from pixel-level flow estimation through shared associations and flow supervision.** Matching operates in coarse token space (top), while dense flow is formed in pixel space (bottom), across both the image features (left) and flow predictions (right). Features from Image 1 and Image 2 are pooled into adaptive tokens and matched across images (blue links). On the right, predicted token flow F_q is combined into pixel flow F_x, while ground-truth pixel flow F_x^{gt} is pooled into pseudo-ground-truth token flow F_q^{gt}. The probabilistic predictions $\mathcal{F}_q$ include token flow F_q and its uncertainty. We learn token matching and pixel-to-token associations using only ground-truth pixel flow, through three complementary losses: $\mathcal{L}_{\mathrm{token}}(\mathcal{F}_q \mid P_q)$ trains token-flow prediction against pooled ground truth, holding P_q fixed in constructing the pseudo-ground truth; $\mathcal{L}_{\mathrm{assign}}(P_x \mid \mathcal{F}_q)$ trains associations to favor tokens that confidently explain pixel motion, holding token-flow predictions fixed; and $\mathcal{L}_{\mathrm{recon}}(P_q, P_x \mid F_x^{\mathrm{gt}})$ trains flow-preserving grouping by pooling and reconstructing fixed ground-truth flow, independently of matching. Upper views enlarge the boxed regions, token colors show pooled RGB, and flow maps share one color scale.

1) Token-flow prediction. Since matching operates at the coarse token level, we construct pseudo-ground-truth token flow by pooling ground-truth pixel flow, provisionally assuming the current associations are correct. Following SEA-RAFT [20], the network predicts token flow F_q and its uncertainty.

We denote these probabilistic predictions by $\mathcal{F}_q$ and define $\mathrm{NLL}(\mathcal{P}; Y) = -\sum_i \log p_{\mathcal{P}_i}(Y_i)$ as the negative log-likelihood of targets Y under predicted distributions $\mathcal{P}$:

$$F_q^{\mathrm{gt}} = P_q F_x^{\mathrm{gt}}, \tag{4}$$

$$\mathcal{L}_{\mathrm{token}}(\mathcal{F}_q \mid P_q) = \mathrm{NLL}(\mathcal{F}_q; F_q^{\mathrm{gt}}). \tag{5}$$

This loss primarily trains token matching, updating token features, local attention, iterative flow refinement, and uncertainty prediction. It also trains pixel-to-token associations through gradients from feature pooling. However, gradients are stopped when constructing pseudo-ground-truth token flow, preventing the associations from shifting this supervision to reduce the loss.

2) Pixel-to-token associations. Near motion boundaries, neighboring tokens may predict different flows. Averaging them with spread-out weights in Eq. (3) can produce a spurious intermediate flow. To preserve distinct motions, we learn pixel-to-token associations based on how well each token's predicted flow explains the pixel's ground truth, accounting for prediction uncertainty. Provisionally holding token-flow predictions and uncertainties fixed, we evaluate the likelihood of each pixel's ground-truth flow under each neighboring token's predicted distribution. The association weights P_x combine these likelihoods into a mixture, denoted by $\mathrm{Mix}_{P_x}[\mathcal{F}_q]$. We minimize

$$\mathcal{L}_{\mathrm{assign}}(P_x \mid \mathcal{F}_q) = \mathrm{NLL}\big(\mathrm{Mix}_{P_x}[\mathcal{F}_q]; F_x^{\mathrm{gt}}\big)\,. \tag{6}$$

Unlike averaging token flows, this mixture retains distinct predictions and uncertainties, encouraging pixels to associate with tokens that confidently explain their motion. With token predictions fixed, this loss updates the assignment head and its input features, not token matching or refinement.

3) Flow-preserving reconstruction. Ground-truth pixel flow should also be preserved when pooled into coarse token flow and reconstructed back at pixel resolution. We therefore minimize the reconstruction error, with an aggregated robust flow penalty $\mathcal{D}$:

$$\mathcal{L}_{\mathrm{recon}}(P_q, P_x \mid F_x^{\mathrm{gt}}) = \mathcal{D}\big(P_x P_q F_x^{\mathrm{gt}}, F_x^{\mathrm{gt}}\big)\,. \tag{7}$$

As in superpixel learning [27], this loss updates the associations through both P_q and P_x, encouraging the shared pixel-to-token associations to preserve distinct pixel motions. Since matching is bypassed, the loss trains grouping independently of token-flow prediction.

4) Joint training. We optimize all three losses jointly, applying token-flow and association losses at each refinement step and reconstruction loss once (Appendix G). Together, they learn token flow and pixel-to-token associations to achieve both reliable matching and precise pixel-level flow.

4 Experiments

Our experiments test whether coupling a correspondence token's matching evidence with the pixel motion it represents improves dense flow prediction. We first compare zero-shot accuracy with published methods. We then examine where errors arise and where FATE reduces them, focusing on two regions: occluded or out-of-frame pixels, which lack a direct match and depend on shared evidence, and pixels near motion boundaries, where distinct motions must be preserved. A comparison with a fixed-grid model at the same token count tests the formulation itself. Ablations test its design choices, and computation profiles show what fewer tokens make affordable.

Experimental setup. All compared methods are trained on FlyingChairs [13] followed by FlyingThings3D (Things) [14] and evaluated on two frames.

1) Models and training. We build FATE on WAFT's iterative matching and refinement framework [8] with a DINOv3 ViT-S backbone [29]. For an $H \times W$ image, both extract dense features on an $(H/2) \times (W/2)$ grid, abbreviated $H/2$. Released WAFT estimates flow on this grid, whereas FATE estimates it for adaptive tokens anchored on an $H/4$ lattice, with a quarter as many tokens. FATE and WAFT use five refinement steps in training and inference unless stated otherwise (Appendix E).

2) Evaluation protocol. Things test results are in distribution, while results on the Sintel [15] and KITTI 2015 [16] training splits are zero-shot, without target-domain flow training, as in real scenes where dense flow labels are difficult to acquire. Things and Sintel each have Clean and Final versions.

3) Metrics. We report endpoint error (EPE), the Euclidean flow error in pixels averaged per image and then over images, and KITTI Fl, the percentage of valid pixels whose error exceeds both 3 pixels and 5% of the ground-truth motion. Regional analyses pool endpoint errors over pixels (Appendix B). Computation is given in GMACs, billions of multiply–accumulates, per 540×960 image pair.

Table 1: **FATE achieves the lowest zero-shot Sintel Final and KITTI EPE among the compared methods.** All methods train on FlyingChairs and FlyingThings3D. Errors are EPE except KITTI Fl. Bold: lowest error in each accuracy column; –: not reported.

Method	Sintel Clean	Sintel Final	KITTI EPE	KITTI Fl (%)	GMACs
FlowNet2 [31]	2.02	3.14	10.10	30.4	230
PWC-Net [17]	2.55	3.93	10.40	33.7	101
RAFT [5]	1.43	2.71	5.04	17.4	938
GMA [22]	1.30	2.74	4.69	17.1	1352
SKFlow [32]	1.22	2.46	4.27	15.5	751
DIP [33]	1.30	2.82	4.29	13.7	3068
CRAFT [34]	1.27	2.79	4.88	17.5	2274
GMFlowNet [35]	1.14	2.71	4.24	15.4	1094
FlowFormer [7]	1.01	2.40	4.09	14.7	1715
EMD-L [36]	0.88	2.55	4.12	13.5	1755
FlowFormer++ [37]	0.90	2.30	3.93	14.1	1713
MatchFlow(G) [18]	1.03	2.45	4.08	15.6	1669
AnyFlow [24]	1.10	2.52	3.76	12.4	–
RPKNet [38]	1.12	2.45	–	13.0	137
CCMR+ [39]	0.98	2.36	–	12.9	12653
SEA-RAFT(L) [20]	1.19	4.11	3.62	12.9	655
SAMFlow [19]	0.87	2.11	3.44	12.3	9717
FlowDiffuser [21]	**0.86**	2.19	3.61	11.8	2466
DPFlow [23]	1.02	2.26	3.37	**11.1**	414
WAFT [8]	1.28	2.56	3.49	12.9	732
FATE	1.03	**2.10**	**3.11**	11.9	796

Table 2: **FATE's gain over WAFT is larger at occluded or out-of-frame pixels and near motion boundaries than at coherent pixels.** The categories partition valid pixels, with visible pixels split by their distance to ground-truth motion boundaries (Appendix B). Pixel and Error give percentages of the pixels and of WAFT's summed endpoint error; WAFT and FATE give pooled EPE, and Gain is WAFT minus FATE. Bold marks the lower EPE in each pair.

	Coherent					Near boundary					Occluded/out-of-frame				
Dataset	Pixel	Error	WAFT	FATE	Gain	Pixel	Error	WAFT	FATE	Gain	Pixel	Error	WAFT	FATE	Gain
Sintel Clean	87.9	36.4	0.52	**0.38**	0.14	3.6	10.8	3.79	**3.24**	0.54	8.4	52.7	7.96	**6.78**	1.18
Sintel Final	87.9	46.7	1.35	**1.05**	0.29	3.6	10.7	7.51	**6.63**	0.87	8.4	42.5	12.88	**10.93**	1.95
KITTI	77.8	36.8	**1.72**	1.76	−0.04	8.1	10.8	4.83	**4.81**	0.01	13.9	52.3	13.63	**11.05**	2.57
Things Clean	84.8	35.4	1.45	**0.92**	0.53	8.7	22.0	8.82	**5.76**	3.06	6.4	42.5	23.04	**16.56**	6.47
Things Final	84.8	35.1	1.39	**0.93**	0.46	8.7	22.9	8.87	**6.03**	2.83	6.4	41.9	21.96	**15.88**	6.07

Zero-shot transfer and regional accuracy. We compare overall accuracy with published results for prior methods and regional accuracy with the released WAFT checkpoint.

1) FATE achieves the lowest zero-shot EPE on Sintel Final and KITTI. In Table 1, FATE has lower Sintel Final and KITTI EPE than every compared method and improves on WAFT in every accuracy column. With a Depth Anything V2 backbone [30] instead of DINOv3, FATE again has lower EPE than WAFT on both Sintel and KITTI (Appendix A).

2) Pixels without direct matches or near motion boundaries carry most of WAFT's error. In Table 2, occluded or out-of-frame pixels make up only 6.4–13.9% of the pixels yet account for 41.9–52.7% of WAFT's summed endpoint error. Near-boundary pixels likewise account for 10.7–22.9% of the error from 3.6–8.7% of the pixels. Coherent pixels, the remaining 77.8–87.9%, account for only 35.1–46.7% of the error, so overall EPE reveals little about accuracy in the other two regions. Fig. 5 illustrates three cases that make flow hard: a thin structure near motion boundaries, an occluded background and misleading lighting. Fig. 6 shows more such cases over wider views.

3) FATE's largest gains occur where direct matches are missing. Coherent pixels gain at most 0.53 pixels, versus 1.18–2.57 pixels for occluded or out-of-frame pixels on Sintel and KITTI and over 6 on Things, the largest gain in each row of Table 2. Without direct matches, these pixels' motion must be inferred from shared evidence, such as visible parts of the same moving region. In Fig. 5(b), FATE recovers the motion of leaves hidden in Image 2, unlike WAFT; Fig. 6 shows similar cases

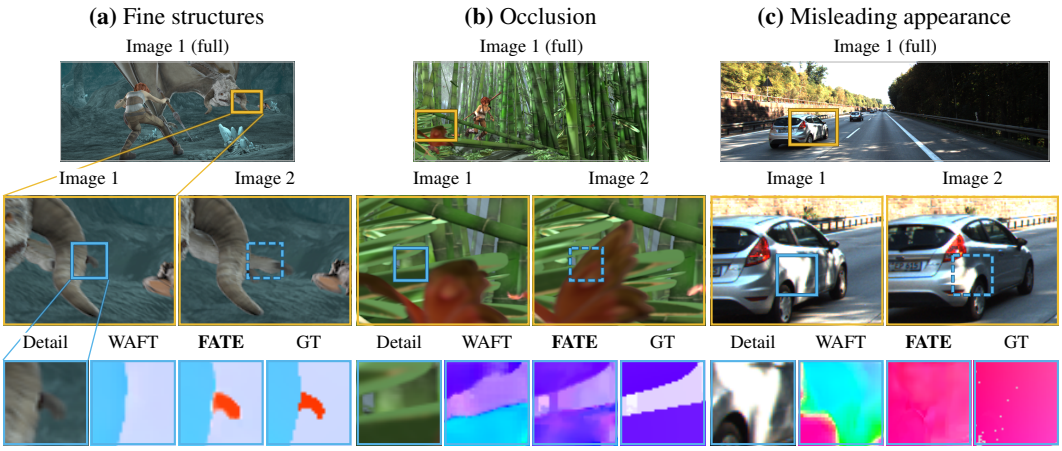


Figure 5: **FATE better handles fine structures, occlusion and misleading appearance.** (a) A thin protrusion moves unlike the surfaces on both sides; per-pixel assignments keep its motion distinct. (b) Leaves hidden by a wing in Image 2 have no match; their tokens infer motion from surrounding context. (c) Sunlight and shadows on a car change between frames; WAFT's flow follows them, while FATE's tokens keep the whole surface's motion coherent. Gold and blue boxes locate context and detail crops; dashed boxes repeat Image 1 coordinates in Image 2. All predictions are zero-shot; colors share a scale per example, and gray marks unlabeled GT.

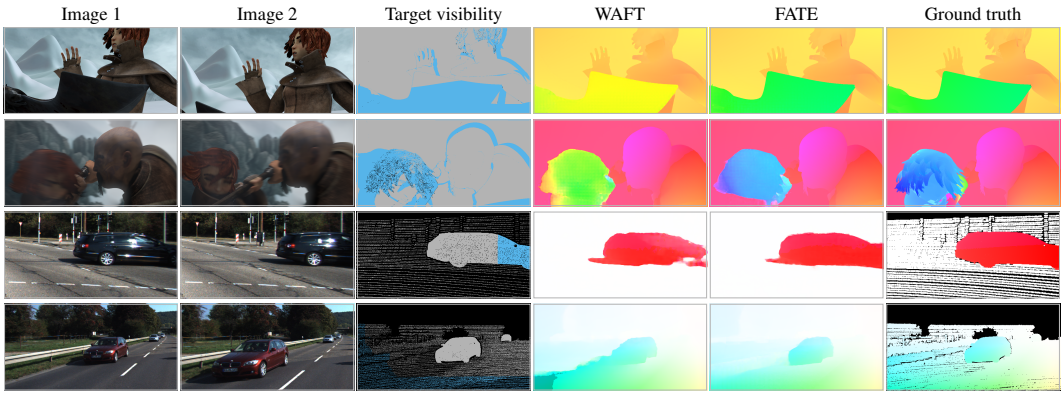


Figure 6: **FATE recovers motion where matches are missing or ambiguous and preserves motion boundaries.** Matches are missing for an occluded cape (row 1) and a car leaving the frame at the right border (row 3) and ambiguous for blurred hair (row 2); an approaching car's motion must stay separate from the road's (row 4). Target visibility marks occluded or out-of-frame pixels in blue, other valid pixels in gray and excluded pixels in black. All predictions are zero-shot, on Sintel (rows 1–2) and KITTI (rows 3–4); each row shares one crop and color scale, and sparse KITTI labels are enlarged for display. Appendix C adds more examples.

in rows 1 and 3. Each token pools evidence over its learned support, draws on local context, and contributes its motion estimate to associated pixels; Table 3(b) tests the benefit of context at equal token counts. Since these pixels are few, overall EPE dilutes their gain to 0.38 pixels on KITTI (Table 1), suggesting greater benefit where missing matches are more common.

4) FATE is more precise near motion boundaries with a quarter of WAFT's tokens. Mixing motions from different surfaces can give boundary pixels incorrect flow, a risk finer tokens should reduce. Yet FATE's near-boundary EPE is 0.54–3.06 pixels lower than WAFT's on Sintel and Things, three to six times its coherent gain, and matches WAFT on KITTI. In Fig. 5(a), WAFT assigns a thin protrusion the background's motion, whereas FATE keeps it distinct. Including occluded or out-of-frame pixels, FATE lowers boundary EPE on every dataset at all three thresholds (Appendix B.2; Fig. 6, row 4). As Figs. 1 and 3 show, learned support follows moving surfaces across grid-cell borders, separating pixels on opposite sides of motion boundaries.

Table 3: **At the same token count, adaptive tokens improve accuracy, gain more from local context and preserve more pixel motion.** All systems estimate flow for $H/4$ tokens. (a) Overall accuracy. (b) Occluded/out-of-frame EPE; attention gain is EPE without minus with local attention. (c) EPE of ground-truth flow pooled into tokens and returned to pixels with each model's image-predicted weights, without retraining. Bold marks the best in (a) and (c) and the larger gain in (b).

(a) Overall accuracy		Things		Sintel		KITTI	
System	Attn.	Clean	Final	Clean	Final	EPE	Fl (%)
WAFT-H/4	–	2.93	2.72	1.08	2.49	3.96	13.3
WAFT-H/4	✓	2.73	2.62	1.07	2.57	3.75	12.6
FATE	–	2.56	2.48	1.12	2.14	3.54	13.0
FATE	✓	**2.39**	**2.38**	**1.03**	**2.10**	**3.11**	**11.9**
(c) Supplied-motion reconstruction EPE							
WAFT-H/4	✓	0.50	0.53	0.21	0.39	1.52	–
FATE	✓	**0.19**	**0.21**	**0.09**	**0.19**	**0.39**	–

(b) Occluded/out-of-frame EPE		Things		Sintel		KITTI
System	Attn.	Clean	Final	Clean	Final	EPE
WAFT-H/4	–	19.65	17.79	7.15	12.91	14.87
WAFT-H/4	✓	18.98	17.64	7.19	12.90	14.09
Attention gain		0.67	0.15	−0.03	0.01	0.78
FATE	–	17.49	16.55	6.89	11.40	12.96
FATE	✓	16.56	15.88	6.78	10.93	11.05
Attention gain		**0.93**	**0.66**	**0.11**	**0.46**	**1.90**

Adaptive versus fixed-grid correspondence. FATE differs from released WAFT in the number of tokens and how they are formed. To remove the first difference, we train WAFT-H/4, a fixed-grid baseline that estimates flow on an $H/4$ grid with WAFT's code and loss. It matches FATE's token count, backbone, dense-feature resolution, iterative framework, refinement steps and training recipe, but its tokens are fixed grid cells. We train both with and without local attention and compare accuracy, the use of context for pixels without direct matches and the preservation of pixel motion.

1) Adaptive tokens improve every reported accuracy result. In Table 3(a), FATE outperforms both WAFT-H/4 variants on every reported accuracy result. With attention in both systems, it lowers Sintel Final EPE from 2.57 to 2.10 and KITTI EPE from 3.75 to 3.11. Without attention in either, FATE still improves every result except Sintel Clean EPE. FATE's advantage over WAFT therefore does not come from using coarser tokens or from local attention alone.

2) Adaptive tokens benefit more from local context. In Table 3(b), FATE has lower occluded/out-of-frame EPE than WAFT-H/4 even without local attention, and adding attention widens this gap on every dataset. Attention lowers FATE's EPE by 1.90 pixels on KITTI versus 0.78 for WAFT-H/4, and changes WAFT-H/4's EPE on Sintel by less than 0.05 pixels. This supports our account of the gains in Table 2: learned support defines which pixels share an estimate, so a token can draw on broader context while its estimate remains tied to the pixels it represents.

3) Learned support preserves more pixel motion. Grouping pixels to share evidence should not erase the motion differences among them. To measure how much pixel motion survives grouping, independent of matching accuracy, we replace predicted token flow with pooled ground truth and return it to pixels with each model's image-predicted weights (Appendix D). In Table 3(c), FATE has lower reconstruction EPE than WAFT-H/4 on every dataset: 0.19 versus 0.39 on Sintel Final and 0.39 versus 1.52 on KITTI. At the same token count, learned support thus preserves motion that fixed cells average away. Consistent with this, FATE also has lower EPE than WAFT-H/4 near motion boundaries on every dataset at all three tested thresholds (Appendix B.2).

Learning effective correspondence tokens. Table 4 tests how supervision, local context and the assignment parameterization help tokens learn, changing one choice at a time and training each with the same recipe. Each design choice improves at least five of the six reported results.

1) Training the grouping to preserve motion improves every result. Compared with training without it, the reconstruction loss (Eq. (7)) improves every result, most on Things, where Clean EPE falls from 2.68 to 2.39. Unlike the token and assignment losses, which act through predicted token flow, it trains pooling and broadcasting directly on ground-truth flow, so the grouping learns which motion differences to preserve independently of matching accuracy.

2) Local attention improves every result. Compared with a learned projection of pooled features, local attention improves every result, most on KITTI, where EPE falls from 3.54 to 3.11. It lets each token draw on nearby dense features, while the assignments still decide which pixels receive its flow.

Table 4: **Each design choice improves at least five of the six results.** Each variant changes one choice of FATE and is trained with the same recipe. *No reconstruction loss* trains without Eq. (7). *No local attention* forms token features by a learned projection of pooled features. *End-to-end assignment supervision* lets the assignment loss (Eq. (6)) also update token flow and uncertainty; pseudo-ground-truth token flow stays fixed. With *independent assignment scores*, the assignment head predicts the nine candidate scores directly instead of through a Gaussian. Errors are EPE except KITTI Fl. Bold marks the lowest error in each column.

Variant	Things Clean	Things Final	Sintel Clean	Sintel Final	KITTI EPE	KITTI Fl (%)
FATE	**2.39**	**2.38**	1.03	**2.10**	**3.11**	**11.9**
No reconstruction loss	2.68	2.61	1.05	2.17	3.14	12.4
No local attention	2.56	2.48	1.12	2.14	3.54	13.0
End-to-end assignment supervision	2.49	2.46	**1.00**	2.18	3.64	12.3
Independent assignment scores	2.58	2.55	1.02	2.24	3.16	12.1

3) Supervising assignments directly, rather than end to end, improves accuracy. FATE supervises each module for its role (Fig. 4). The token loss trains matching to produce each token's correspondence, and the assignment loss trains the assignments to decide which token should carry each pixel, given the current probabilistic token-flow predictions. Applied end to end, the assignment loss also updates these predictions, so it can be lowered by shifting a token's flow or widening its uncertainty toward a disagreeing pixel instead of reassigning that pixel. Direct assignment supervision improves every result except Sintel Clean EPE, most on KITTI, where EPE falls from 3.64 to 3.11. Features, token predictions and assignments are still learned jointly.

4) Gaussian assignment scores outperform independently predicted scores. Compared with predicting the nine candidate scores independently from the same head, Gaussian scores improve every result except Sintel Clean EPE, including Sintel Final EPE, which falls from 2.24 to 2.10. The Gaussian couples each pixel's candidate scores through the anchors' relative positions (Appendix G), yet a token's support can still take an irregular shape (Fig. 3).

Accuracy and computation. FATE predicts assignments at image resolution but matches and refines fewer tokens. Appendix E profiles how this allocation affects accuracy and cost.

1) Fewer tokens make wider refinement and local attention affordable. In Table 7, with a quarter as many tokens, FATE refines with twice WAFT's recurrent-state and refinement width at similar refinement cost, 453.9 versus 425.8 GMACs, whereas widening WAFT the same way would more than double its refinement cost. Local attention of the same width likewise costs 20.6 GMACs over FATE's $H/4$ tokens versus 57.0 GMACs over WAFT's $H/2$ grid.

2) With four refinement steps, FATE outperforms five-step WAFT at lower cost. In Table 8, four-step FATE has lower error than five-step WAFT on every reported result while using 706.1 versus 732.4 GMACs. Our five-step default costs more than WAFT, 796.8 GMACs, and further lowers error on five of the six reported results. A narrower FATE with WAFT's width costs 636.3 GMACs and has lower error than WAFT on every reported result except KITTI EPE.

5 Conclusion

Optical flow requires reliable matching from sufficient visual context and precise motion at every pixel. FATE reconciles these demands through consistent visual grounding: pixels likely to move together share visual evidence for matching and draw on the same motion estimates, so correspondence can operate at a coarser scale than the flow. Shared pixel-to-token associations pool features into adaptive tokens for matching and combine token motions into dense flow. Flow supervision learns these associations without grouping labels, allowing the model to predict them from appearance alone.

FATE achieves the lowest zero-shot Sintel Final and KITTI EPE among compared published methods. Its largest gains occur at occluded or out-of-frame pixels, with further gains near motion boundaries. At equal token counts, adaptive grouping outperforms fixed-grid alternatives, benefits more from local attention, and better preserves pixel motion. With four refinement steps, FATE also outperforms five-step WAFT at lower cost. These results show that learned pixel-to-token associations enable both reliable coarse-scale matching and precise pixel-level flow, generalizing well to unseen real scenes.

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Common setting. Unless stated otherwise, we evaluate the DINOv3 systems from Section 4: FATE, released WAFT, and the fixed-grid WAFT-H/4 baseline with local attention, all trained on FlyingChairs and FlyingThings3D. The fine-tuned results in Appendix F are the exception.

A Zero-Shot Transfer across Backbones

In Table 5, FATE lowers zero-shot EPE on Sintel and KITTI with both DINOv3 and Depth Anything V2 (DAv2) [30], compared with WAFT's published results [8]. KITTI Fl also improves with DINOv3, and with DAv2 the two models tie at the reported precision.

Table 5: **FATE lowers EPE with both backbones.** Zero-shot results on the Sintel and KITTI training splits. Bold: lowest error within each backbone, including ties at the reported precision.

		Sintel		KITTI	
Backbone	Method	Clean	Final	EPE	Fl (%)
DINOv3	WAFT [8]	1.28	2.56	3.49	12.9
	FATE	**1.03**	**2.10**	**3.11**	**11.9**
DAv2	WAFT [8]	1.01	2.49	3.28	**10.9**
	FATE	**0.97**	**2.30**	**2.93**	**10.9**

B Regional Evaluation

The regional analyses test accuracy where direct matches are unavailable, at occluded or out-of-frame pixels, and where neighboring motions differ, near motion boundaries.

B.1 Occluded and out-of-frame pixels

Pixel sets. A pixel is out of frame if its ground-truth endpoint $x + F_x^{\mathrm{gt}}(x)$ falls outside Image 2. On Sintel and KITTI, we combine these pixels with those whose targets are marked occluded by the dataset annotations. On Things, we use endpoint bounds only, so in-frame occlusions are not identified and visible means in frame. All sets are restricted to the valid pixels used for standard EPE.

Sintel's official invalid mask further excludes pixels from visibility analysis, even when they remain valid for standard EPE. Table 2 partitions the remaining population: visible-boundary pixels lie in the boundary neighborhoods defined below, visible-coherent pixels are their complement among visible pixels (coherent does not imply constant flow), and the third group is non-visible.

Regional EPE and error share. Let V be the pixels eligible for visibility analysis, pooled over all evaluated image pairs, $R \subseteq V$ one of the three categories, and $e(x) = \|F_x(x) - F_x^{\mathrm{gt}}(x)\|_2$ the endpoint error at pixel x. Table 2 reports the regional EPE and, as a percentage, WAFT's error share

$$\mathrm{EPE}_R = \frac{1}{|R|} \sum_{x \in R} e(x), \qquad S_R = \frac{\sum_{x \in R} e(x)}{\sum_{x \in V} e(x)}. \tag{8}$$

Unlike standard EPE, which averages within each image first, both weight every pixel equally. The pixel and error shares of the three categories therefore each sum to 100% before truncation.

B.2 Accuracy near motion boundaries

Defining the evaluation region. To test accuracy where neighboring motions differ, we measure EPE over a region we define around ground-truth motion boundaries. We identify these boundaries from local changes in ground-truth flow F_x^{gt}, with strength

$$s(x) = \|\Delta_1 F_x^{\mathrm{gt}}(x)\|_2 + \|\Delta_2 F_x^{\mathrm{gt}}(x)\|_2, \tag{9}$$

where each Δ takes the difference to the next pixel along one image axis. A difference contributes only when both pixels have valid labels. For threshold τ, pixels with $s(x) > \tau$ seed the boundary

region, which contains valid pixels in a 5×5 window around any seed. We pool endpoint errors over this region. We evaluate $\tau = 2, 3, 4$ pixels; Table 2 uses $\tau = 2$ intersected with the visible set on Sintel and KITTI, or the in-frame set on Things. The threshold study in Table 6 includes both visible and non-visible pixels. At $\tau = 2$, this full boundary region covers 9.3% of valid pixels on Things, 6.2% on Sintel and 13.6% on KITTI.

Boundary precision improves across the tested thresholds. In Table 6, FATE has lower EPE in this region than both fixed-grid models on every dataset at all three thresholds.

Table 6: **FATE has lower EPE near motion boundaries at all tested thresholds.** EPE is pooled over the boundary region, including non-visible pixels; the threshold sets how large a local flow change must be to count as a boundary. WAFT estimates motion for four times as many tokens as FATE, and WAFT-H/4 with attention for the same number.

	Things		Sintel		KITTI
System	Clean	Final	Clean	Final	EPE
(a) Boundary threshold: 2 pixels					
WAFT	13.40	13.14	7.30	11.92	9.97
WAFT-H/4 + attention	10.77	10.65	6.97	11.40	11.75
FATE	**9.24**	**9.48**	**6.63**	**10.45**	**8.51**
(b) Boundary threshold: 3 pixels					
WAFT	12.50	12.37	7.38	12.52	12.84
WAFT-H/4 + attention	10.08	10.07	7.02	11.90	15.30
FATE	**8.56**	**8.77**	**6.64**	**11.03**	**10.65**
(c) Boundary threshold: 4 pixels					
WAFT	12.29	12.26	7.75	13.38	15.75
WAFT-H/4 + attention	10.02	9.99	7.34	12.73	18.87
FATE	**8.40**	**8.63**	**6.85**	**11.78**	**12.76**

C Qualitative Flow and Learned Support

Fig. 7 adds two comparisons with clear improvements over WAFT, using the zero-shot DINOv3 models from Fig. 6. Each pairs the flow comparison with learned assignments in a boxed region. These selected examples illustrate improvements; the tables assess performance across datasets.

Reading the assignments. White contours separate pixels whose strongest assignments select different tokens; the model itself uses soft weights. Assignment entropy, normalized by $\log 9$, ranges from dark blue (weight concentrated on one token) to yellow (weight spread across tokens). It describes assignment spread, not flow confidence.

Visualization protocol. Here and in Fig. 6, ground truth and both predictions share the Middlebury color wheel within each example. On Sintel, magnitude m uses radius $\log(1 + m)/6$; on KITTI, radius grows linearly up to the largest valid ground-truth magnitude, and black marks missing labels. Target-visibility colors are as defined in the caption of Fig. 6.

D Supplied-Motion Reconstruction Protocol

This diagnostic measures how much pixel motion each model preserves when flow is pooled into tokens and returned to pixels, independent of its token-flow accuracy. Table 3(c) reports the results.

Substituting pooled ground-truth flow. We first predict each model's weights from the input images and keep them fixed; neither model is retrained. For FATE, P_q pools ground-truth flow into tokens and P_x broadcasts it back through the same assignments A. This is the ground-truth round trip used by reconstruction supervision (Eq. (7)) and illustrated in Fig. 4. For WAFT-H/4 with local attention, we average ground-truth flow within each fixed cell and upsample it with the predicted

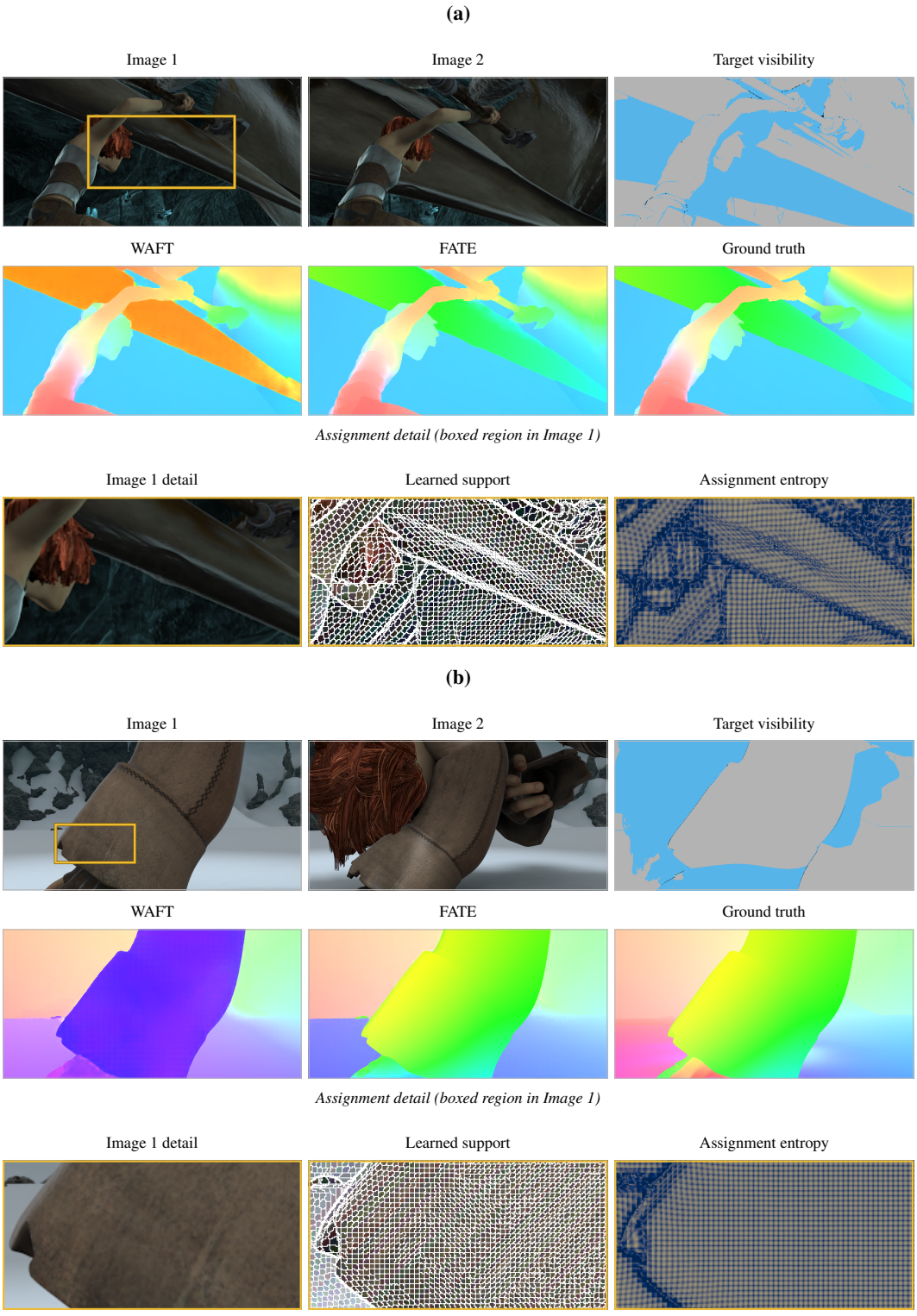


Figure 7: **FATE recovers motion on both thin and broad structures.** (a) WAFT assigns incorrect motion to the wing behind the crossing limbs; FATE recovers the wing's motion. (b) FATE recovers the sleeve's spatially varying motion. Both are Sintel Clean pairs. Gold-framed details show the learned support and assignment entropy.

convex weights. Pooling and evaluation use valid ground-truth pixels. We report reconstruction EPE, averaged within each image and then over images, rather than the robust training penalty $\mathcal{D}$ in Eq. (7).

What the reconstruction measures. Lower reconstruction EPE means that less pixel motion is lost in the round trip. The comparison evaluates pooling and the return to pixels together; it does not isolate their effects or decompose the end-to-end prediction error.

E Accuracy and Computation

Computation is measured per 540×960 image pair with model-specific padding. Counts include feature extraction, token matching features, all refinement steps and the final per-pixel flow output.

Fewer tokens make wider refinement and local attention affordable. In Table 7, suffixes give recurrent-state and refinement widths: WAFT-64 is released WAFT, and FATE-128 is our main model. At WAFT's width, refinement over FATE's tokens costs 293.8 versus 425.8 GMACs. Widening WAFT to FATE-128's width, profiled without training, raises its refinement cost to 943.9 GMACs.

Table 7: **Fewer tokens make wider refinement and local attention affordable.** Resolution denotes the token lattice; suffixes give recurrent-state and refinement widths; Steps counts refinement steps. Costs are GMACs per 540×960 image pair; errors are EPE except KITTI Fl. Bold marks the lowest refinement cost, total cost and error in each column. †: profiled only, not trained.

			Cost (GMACs)			Things		Sintel		KITTI	
System	Resolution	Steps	Attention	Refinement	Total	Clean	Final	Clean	Final	EPE	Fl (%)
WAFT-64 [8]	$H/2$	5	–	425.4	732.4	3.54	3.44	1.28	2.56	3.49	12.9
WAFT-128 + attn.†	$H/2$	5	57.0	943.9	1313.2	–	–	–	–	–	–
FATE-64	$H/4$	5	20.6	**293.8**	**636.3**	2.55	2.48	**1.03**	2.24	3.52	12.5
FATE-128	$H/4$	4	20.6	363.1	706.1	2.44	2.43	**1.03**	2.15	3.18	12.2
FATE-128 (main)	$H/4$	5	20.6	453.9	796.8	**2.39**	**2.38**	**1.03**	**2.10**	**3.11**	**11.9**

Fewer refinement steps retain a favorable accuracy–computation trade-off. Table 8 evaluates FATE and WAFT, both trained with five refinement steps, with one to five steps at inference and no retraining. At the same number of steps, FATE has lower error than WAFT on every reported result but uses more computation. Three-step FATE improves on five-step WAFT on every reported result except KITTI Fl, at 16% lower computation (615.3 versus 732.4 GMACs).

Table 8: **FATE has lower error than WAFT at every number of refinement steps.** Both are trained with five steps and evaluated with one to five without retraining. FATE-64 is separately trained with 64-dimensional recurrent-state and refinement features. Errors are EPE except KITTI Fl.

		Things		Sintel		KITTI		
System	Steps	Clean	Final	Clean	Final	EPE	Fl (%)	GMACs
WAFT	1	6.70	6.44	2.20	3.50	8.47	37.1	391.8
FATE	1	5.01	4.84	1.54	2.67	8.30	36.9	433.8
WAFT	2	4.27	4.15	1.49	2.88	4.67	18.5	477.0
FATE	2	2.91	2.84	1.10	2.29	4.09	17.1	524.5
WAFT	3	3.78	3.66	1.34	2.70	3.81	14.0	562.1
FATE	3	2.55	2.49	1.06	2.22	3.38	13.2	615.3
WAFT	4	3.61	3.51	1.29	2.56	3.56	13.1	647.3
FATE	4	2.44	2.43	1.03	2.15	3.18	12.2	706.1
WAFT	5	3.54	3.44	1.28	2.56	3.49	12.9	732.4
FATE	5	2.39	2.38	1.03	2.10	3.11	11.9	796.8
FATE-64	5	2.55	2.48	1.03	2.24	3.52	12.5	636.3

F Target-Specific Fine-Tuning

The zero-shot advantage does not carry over to the current fine-tuned models. Table 9 compares FATE, fine-tuned with WAFT's recipe and submitted to the benchmark servers, with WAFT's published DINOv3 results [8] on the hidden test sets of Sintel, KITTI 2015, and Spring. FATE matches WAFT's reported Sintel Final EPE and trails it on Sintel Clean, both KITTI Fl variants, and all four Spring metrics. Extending the transfer gains to target-specific fine-tuning remains open.

Benchmark metrics. Sintel reports EPE. KITTI reports Fl over all labeled pixels (Fl-all) and non-occluded pixels (Fl-noc). Spring provides four ground-truth values per image pixel and measures error against the closest one. Its 1px metric is the percentage of pixels with endpoint error above one pixel, and WAUC summarizes accuracy across error thresholds.

Table 9: **After fine-tuning, FATE matches WAFT on Sintel Final but trails it elsewhere.** Benchmark test results for FATE and DINOv3 WAFT. Lower is better except WAUC (↑).

	Sintel		KITTI 2015		Spring			
System	Clean	Final	Fl-all (%)	Fl-noc (%)	1px (%)	EPE	Fl (%)	WAUC ↑
WAFT	0.94	2.02	3.56	2.13	3.18	0.33	1.25	95.05
FATE	1.06	2.02	3.67	2.35	3.43	0.35	1.30	94.86

G Assignment Parameterization and Training Objectives

We specify the assignment scores and training objectives summarized in Section 3. The operators and flow fields retain the same pixel/token subscripts. Matrix entries select an individual pixel or token and flow component; for example, $[F_q]_{q,c}$ is component c of token q's flow. Both transformations in the loss signatures are determined by the shared assignments A.

Gaussian assignment scores. For pixel x, the anchor of its lattice cell defines the local origin. Its candidate anchors $q \in \mathcal{N}(x)$ have offsets $\delta_{xq} \in \{-1, 0, 1\}^2$. The five predicted parameters are a mean offset $\mu_x \in \mathbb{R}^2$ and three entries of a lower-triangular factor L_x with positive diagonal, which defines the inverse covariance $G_x = L_x L_x^\top$. The scores and normalized assignments are

$$a_{xq} = -\tfrac{1}{2}(\delta_{xq} - \mu_x)^\top G_x (\delta_{xq} - \mu_x), \qquad A_{xq} = \frac{\exp a_{xq}}{\sum_{r \in \mathcal{N}(x)} \exp a_{xr}}. \tag{10}$$

Entries outside $\mathcal{N}(x)$ are zero. Since a_{xq} is the log-density of a Gaussian with mean μ_x and covariance G_x^{-1} up to a constant that cancels in the softmax, A_{xq} is that density at δ_{xq}, normalized over $\mathcal{N}(x)$. The parameterization couples each pixel's candidate scores through their relative spatial positions; it does not constrain a token's collected support to a Gaussian shape.

Probabilistic token-flow predictions. The probabilistic flow field $\mathcal{F}_q^{(t)}$ comprises a distribution for each flow component of each token. We write its density as $p_{q,c}^{(t)}(f)$. Following SEA-RAFT [20], it is a two-component Laplace mixture centered on the corresponding flow estimate:

$$p_{q,c}^{(t)}(f) = \sum_{k=1}^{2} \pi_{q,k}^{(t)} \frac{1}{2b_{q,k}^{(t)}} \exp\left(-\frac{|f - [F_q^{(t)}]_{q,c}|}{b_{q,k}^{(t)}} \right). \tag{11}$$

Here $c \in \{1, 2\}$ indexes the flow components, and $\pi_{q,k}^{(t)}$ are softmax-normalized mixture weights. Both mixture components share the flow center. The implementation bounds the first log-scale to $[0, 10]$ and fixes the second to zero; scales and mixture weights are shared across the two flow components. These mixture weights are distinct from pixel-to-token assignments. Image inputs and model parameters are suppressed in the notation.

Valid labels and pseudo-ground truth. Let V denote the valid training pixels. Write $P_q^V(A)$ for weighted pooling restricted to valid labels. The pseudo-ground-truth token flow is $F_q^{\mathrm{gt}} = P_q^V(A)F_x^{\mathrm{gt}}$, and the validity weight $w_q = [P_q(A)\mathbf{1}_V]_q$ is the fraction of a token's support with valid labels. The pseudo-ground truth and validity weights are recomputed from the current assignments and held fixed for token supervision. Feature pooling and matching remain differentiable through the assignments.

Mixtures of neighboring flow predictions. The pixelwise mixture $\mathrm{Mix}_{P_x}[\mathcal{F}_q^{(t)}]$ in Eq. (6) has component density $p_{x,c}^{(t)}$. Since $[P_x]_{xq} = A_{xq}$, its weights come from the pixel assignments. The assignment objective retains the K largest weights for each pixel and renormalizes them over that set, $\mathcal{S}_K(x)$; with $K = 9$, the mixture uses all candidate tokens:

$$p_{x,c}^{(t)}(f) = \sum_{q \in \mathcal{S}_K(x)} \frac{A_{xq}}{\sum_{r \in \mathcal{S}_K(x)} A_{xr}} \, p_{q,c}^{(t)}(f). \tag{12}$$

Token-flow and uncertainty predictions are held fixed for this objective; the assignments and their shared features learn through the mixture weights. The end-to-end assignment supervision variant in Table 4 removes this restriction while keeping the pseudo-ground-truth token flow fixed. Mixing is performed separately for the two flow components, rather than forming a likelihood around an averaged flow vector. Dense inference still uses all nine assignment weights in $F_x = P_x F_q$.

Loss reductions and joint training. The NLL operator in Eqs. (5) and (6) takes a predicted distribution first and a target second. It averages over flow components and aggregates over tokens or valid pixels. With pseudo-ground truth and validity weights fixed for token supervision, and token predictions fixed for assignment supervision, the two objectives are

$$\mathcal{L}_{\mathrm{token}}^{(t)} = -\frac{1}{2\sum_q w_q} \sum_q w_q \sum_{c=1}^{2} \log p_{q,c}^{(t)}\big([F_q^{\mathrm{gt}}]_{q,c}\big)\,, \tag{13}$$

$$\mathcal{L}_{\mathrm{assign}}^{(t)} = -\frac{1}{2|V|} \sum_{x \in V} \sum_{c=1}^{2} \log p_{x,c}^{(t)}\big([F_x^{\mathrm{gt}}]_{x,c}\big)\,. \tag{14}$$

For motion preservation, we recompute $\widetilde{F}_x^{\mathrm{gt}} = P_x(A)P_q^V(A)F_x^{\mathrm{gt}}$ with both transformations differentiable. With $\rho(z) = \sqrt{\|z\|_2^2 + 10^{-6}} - 10^{-3}$, the reconstruction objective, specifying $\mathcal{D}$ in Eq. (7), is

$$\mathcal{L}_{\mathrm{recon}} = \frac{1}{|V|} \sum_{x \in V} \rho\Big([\widetilde{F}_x^{\mathrm{gt}}]_{x,:} - [F_x^{\mathrm{gt}}]_{x,:}\Big)\,. \tag{15}$$

This branch does not reuse the fixed pseudo-ground-truth token flow. Training numerators and denominators pool entries across the minibatch; these training reductions are distinct from benchmark evaluation conventions.

The first two objectives supervise every refinement step, while reconstruction is added once:

$$\mathcal{L} = \sum_{t=1}^{T} \gamma^{T-t} \left(\mathcal{L}_{\mathrm{token}}^{(t)} + \mathcal{L}_{\mathrm{assign}}^{(t)}\right) + \lambda_{\mathrm{recon}}\mathcal{L}_{\mathrm{recon}}. \tag{16}$$

We use $T = 5$, $\gamma = 0.85$, and $\lambda_{\mathrm{recon}} = 1$. All objectives are optimized jointly. Holding the stated predictions fixed within one objective separates its direct gradient paths; it does not freeze the shared features or introduce alternating optimization.